

Lucian Medrihan

Software Engineer

About Me

Engineer with 5+ years building production systems for public institutions. I care about code that actually ships and scales, from satellite imagery pipelines to offline mobile games. I have contributed to projects for SIAN and ESA, demonstrating strong problem-solving capabilities.

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Skills

Technical Skills: Proficient in **Python**, JavaScript, and Node.js, with extensive experience in **React.js**, **Angular.js**, Flask, and FastAPI. Expertise in **full-stack development** using modern web technologies, complemented by strong skills in **Docker**, **AWS**, and cloud solutions.

Additional Skills: **GIS**, **Machine Learning**, **LLMs**, AI-assisted development, Image Processing, Tailwind CSS, Bootstrap, Openlayers, Linux commands, **Cybersecurity**, Geopandas, JQuery, network programming.

Soft Skills: Proven ability to solve **complex problems**, manage projects effectively, and **lead teams**. Strong time management, critical thinking, and attention to detail.

Education

● **Degree in Computer and Automation Engineering**
Sapienza University of Rome

Languages

Italian	★ ★ ★ ★ ★
Romanian	★ ★ ★ ★ ★
English	★ ★ ★ ★ ★

Achievements & Activities

- **Bug Bounty Hunter:** Found 2 critical RCEs (CVSS 10.0) on major Italian ISP via Cyberdart.eu (\$5k bounties)
- **Google Workshop Winner** for Uni-verCity, presented at Google Tel Aviv
- Co-Founder of **Hackappatoi**, Sapienza's top ethical hacking / CTF team

Experiences

● **Software Engineer**
PR.EL.CART. EUROPE ENGINEERING S.R.L.
Rome, Italy
Jun 2025 - Present

- Developing advanced software solutions in collaboration with the Italian Ministry of Agriculture
- Managing and mentoring a team of 3 junior developers, guiding them in software architecture
- Independently developed the official **AGEA Vinitaly 2026** application using PMTiles technology
- Engineered an AI pipeline on Google Cloud Vertex AI to generate 1m satellite photos from 10m Sentinel imagery, reducing costs from €4/5M to €500/month

● **Software Engineer**
Tiuke S.R.L.
Rome, Italy
Oct 2019 - Jun 2025

- Developed 5+ web and mobile applications for the Italian Ministry of Agriculture, serving over 10,000 active users
- Collaborated in a high-performing 5-member development team with shared decision-making, optimizing releases and task coordination
- Refactored legacy monolithic applications into responsive web apps, reducing page load times by 40%
- Created automation tools for integrating mapping layers, accelerating development time by 3x
- Developed backend systems for automated report generation, saving 15 hours of manual work per week
- Implemented specialized LLM agents for agricultural data extraction, achieving 95% accuracy
- Enhanced training of geospatial segmentation models, increasing parcel detection precision by 20%
- Wrote batch scripts to process terabytes of data, reducing pipeline execution time by 50%

Projects

[View more on GitHub](#)

Card Scanner [GitHub: ★ 58] *Completed*
AI-powered Pokemon card recognition system with client-side processing

- Built a lightweight (<10MB) client-side card identification system that works with all new cards without requiring retraining
- Successfully integrated as the core MVP scanner into the open-source **TCG Pocket Collection Tracker** (★ 150+), significantly enhancing the user experience

EGNSS Capacitor *Completed*
Universal GNSS Capacitor plugin with OSNMA support

- Developed a cross-platform plugin to guarantee absolute geographic precision in Europe using Galileo's OSNMA and EGNOS
- Exposed a unified API for Android, iOS, and Web to read highly accurate coordinates from internal GNSS and external Bluetooth antennas

Wolf & Mafia *Completed*
100% Offline Mafia & Werewolf game for Android and iOS

- Developed from scratch a complete offline social deduction game featuring 36 unique roles and a fully guided narrator system
- Powered by a completely deterministic, seed-based game state generator that functions offline without any internet or servers
- Implemented QR-code based local multiplayer allowing users to join games instantly

Game Ecosystem Tools *In Progress*
ForgeMaster App & Eatventure Seed Predictor

- Developed an auto-updating companion app for ForgeMaster (🌐 / 📱)
- Created an accurate seed-based RNG drop predictor for Eatventure (🌐 / 📱)

Smart House *Completed*
Arduino-based custom home automation system

- Developed a complete home automation system on Arduino MEGA2560 featuring SSH remote access
- Implemented PWM control for lighting, ADC sensor inputs, and a custom interactive shell interface

UniverCity *Completed*
Award-winning collaborative platform for university students

- Created an intelligent note-sharing system with advanced indexing algorithms, winning the Google Workshop competition
- Presented the project at Google's offices in Tel Aviv, receiving recognition for innovation and practical application